



For Performance Measurement

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

COMBINED SCIENCE

PAPER 1 Multiple Choice

4003/1

1 hour

JUNE 2025 SESSION

Additional materials:

Multiple Choice answer sheet

Soft clean eraser

Soft pencil (type B or HB is recommended)

Calculator (optional)

Time: 1 Hour

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your name, centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

Answer **all** questions. There are **forty** questions in this paper.

For each question, there are four possible answers, **A, B, C and D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

Each correct answer will score **one** mark. A mark will **not** be deducted for a wrong answer. Any rough working should be done in this booklet.

This question paper consists of 16 printed pages.

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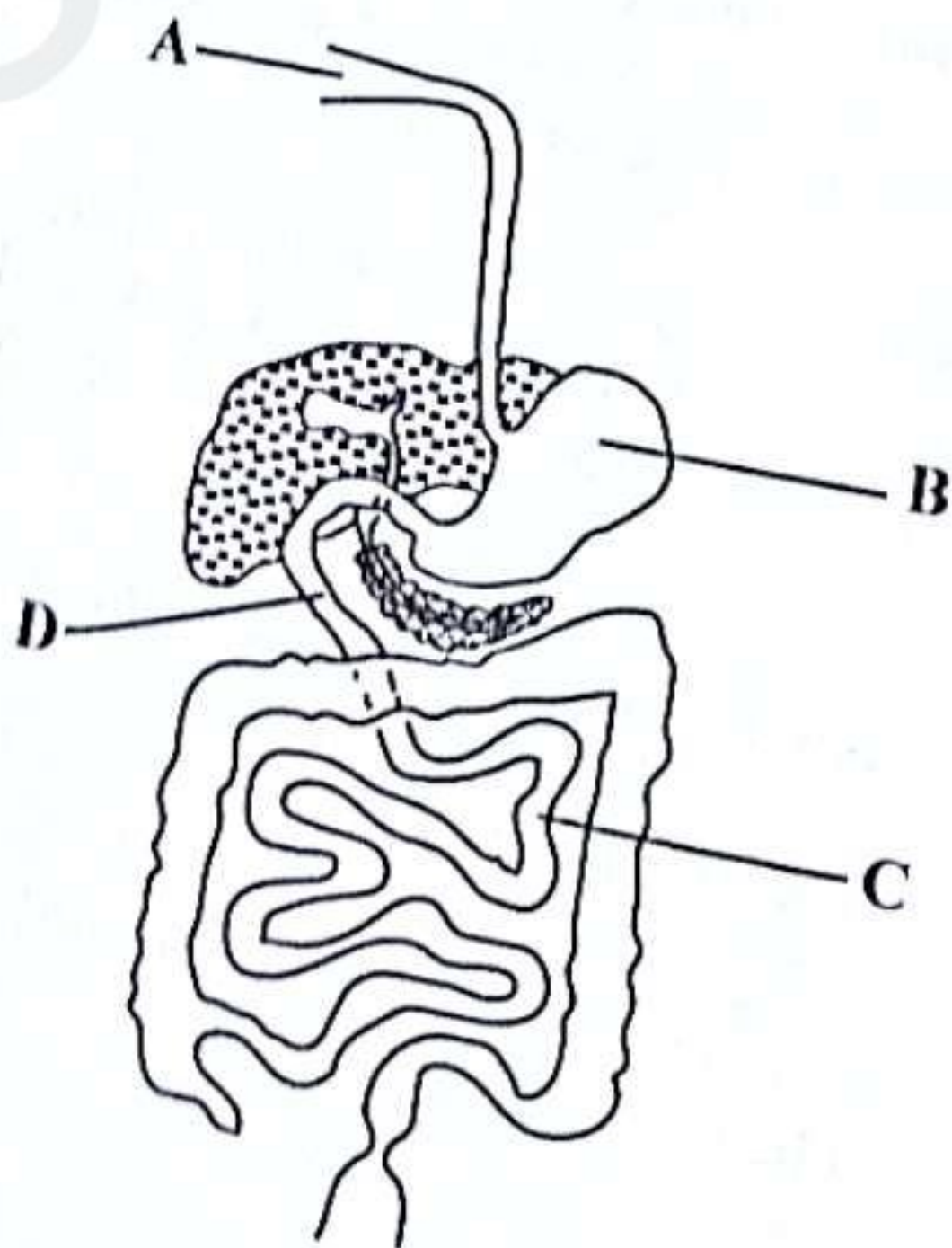
1. Which structure is found in both plant and animal cells?

- A chloroplast
- B nucleus
- C starch grain
- D cellulose cell wall

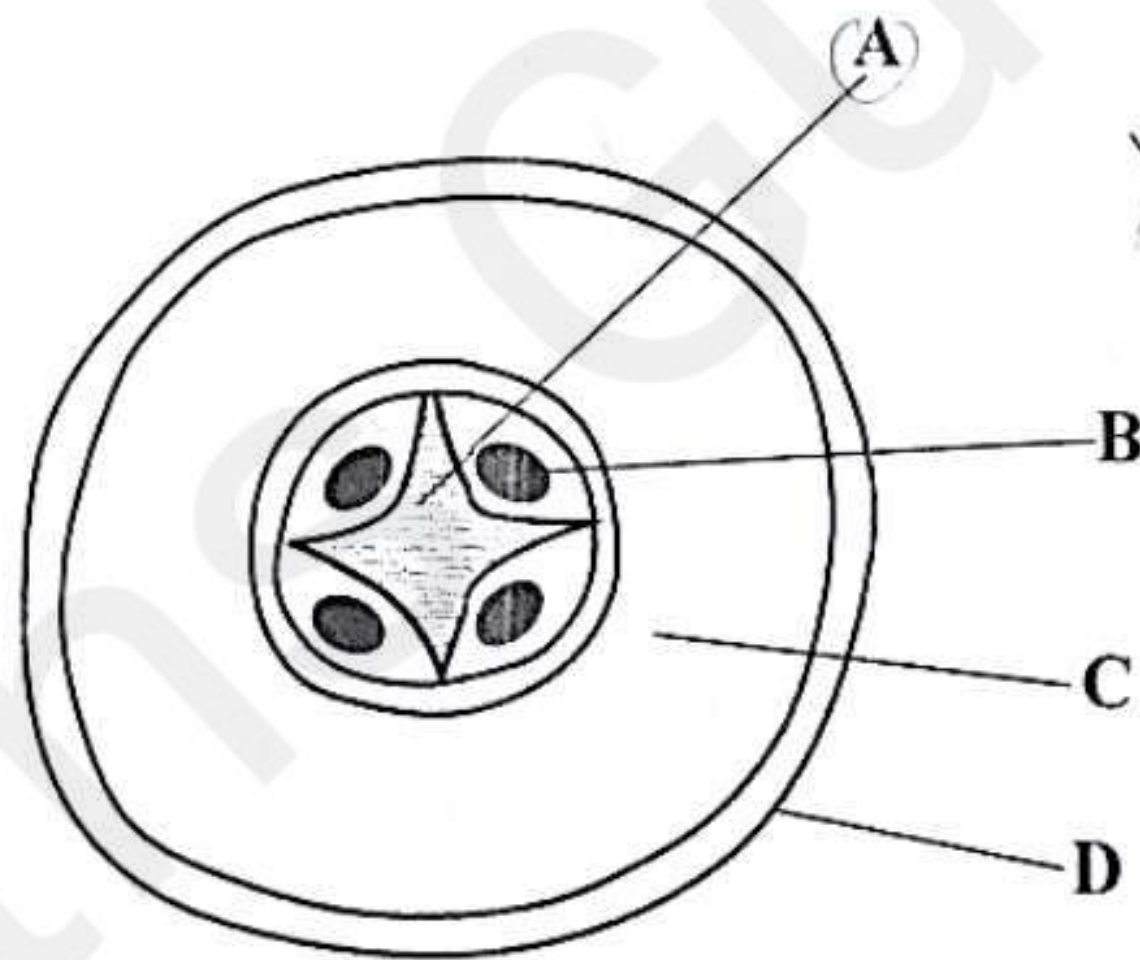
2. Where does the most transpiration take place in a plant?

- A cuticle
- B root hairs
- C stomata
- D xylem

3. The diagram shows a sketch of the human alimentary canal. Where does the chemical digestion of proteins begin?

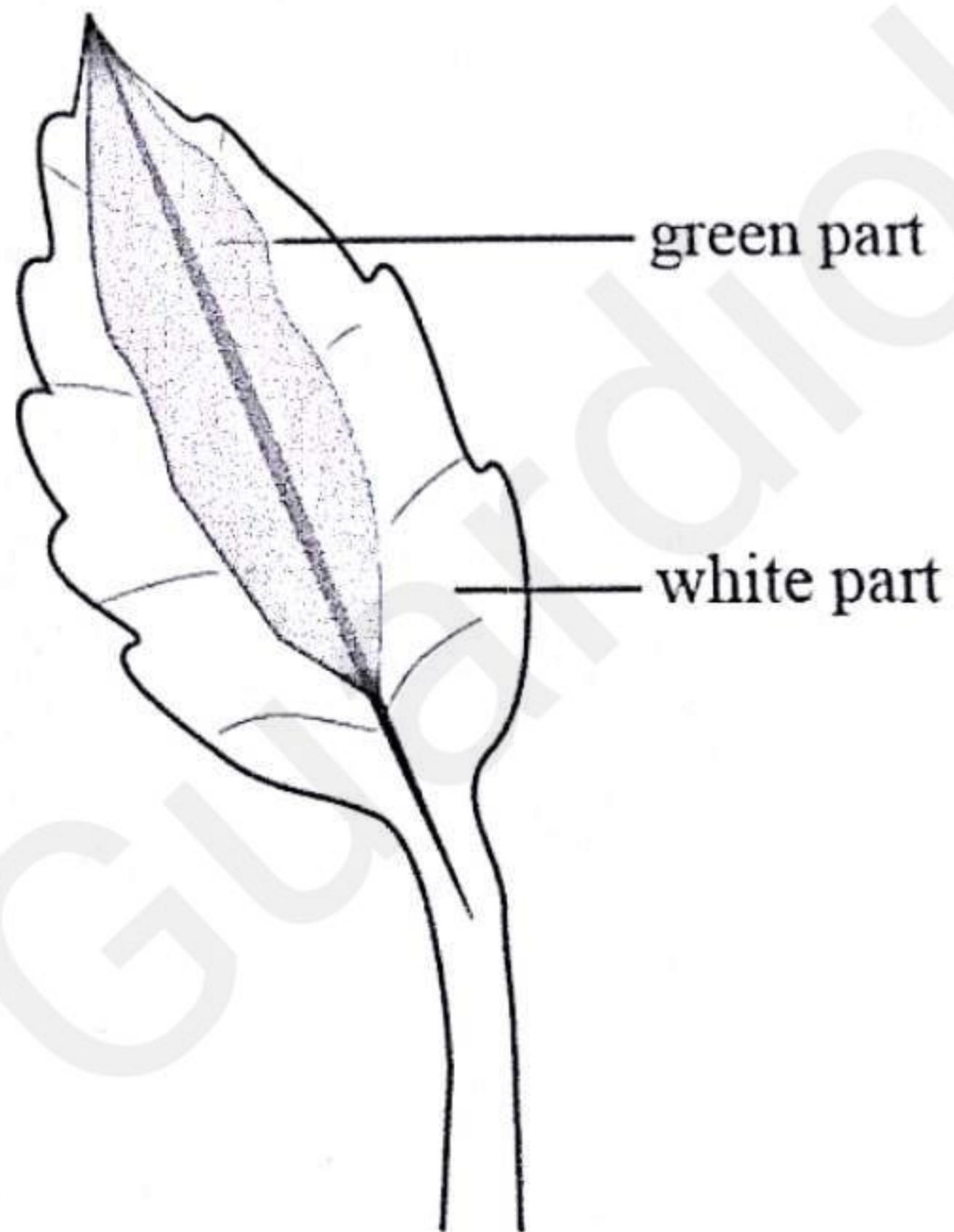


4. What is the function of the gall bladder?
- A production of bile
 - B absorption of fats
 - C digestion of fats
 - D storage of bile
5. Which disease is caused by a deficiency of vitamin A?
- A scurvy
 - B rickets
 - C anaemia
 - D night blindness
6. The diagram shows a transverse section of a dicotyledonous root. Which part transports water?



- Which word equation shows the raw materials and products of photosynthesis?
- A glucose + water $\longrightarrow$ carbon dioxide + water
 - B carbon dioxide + water $\longrightarrow$ glucose + oxygen
 - C glucose + oxygen $\longrightarrow$ carbon dioxide + water
 - D carbon dioxide + glucose $\longrightarrow$ water + oxygen

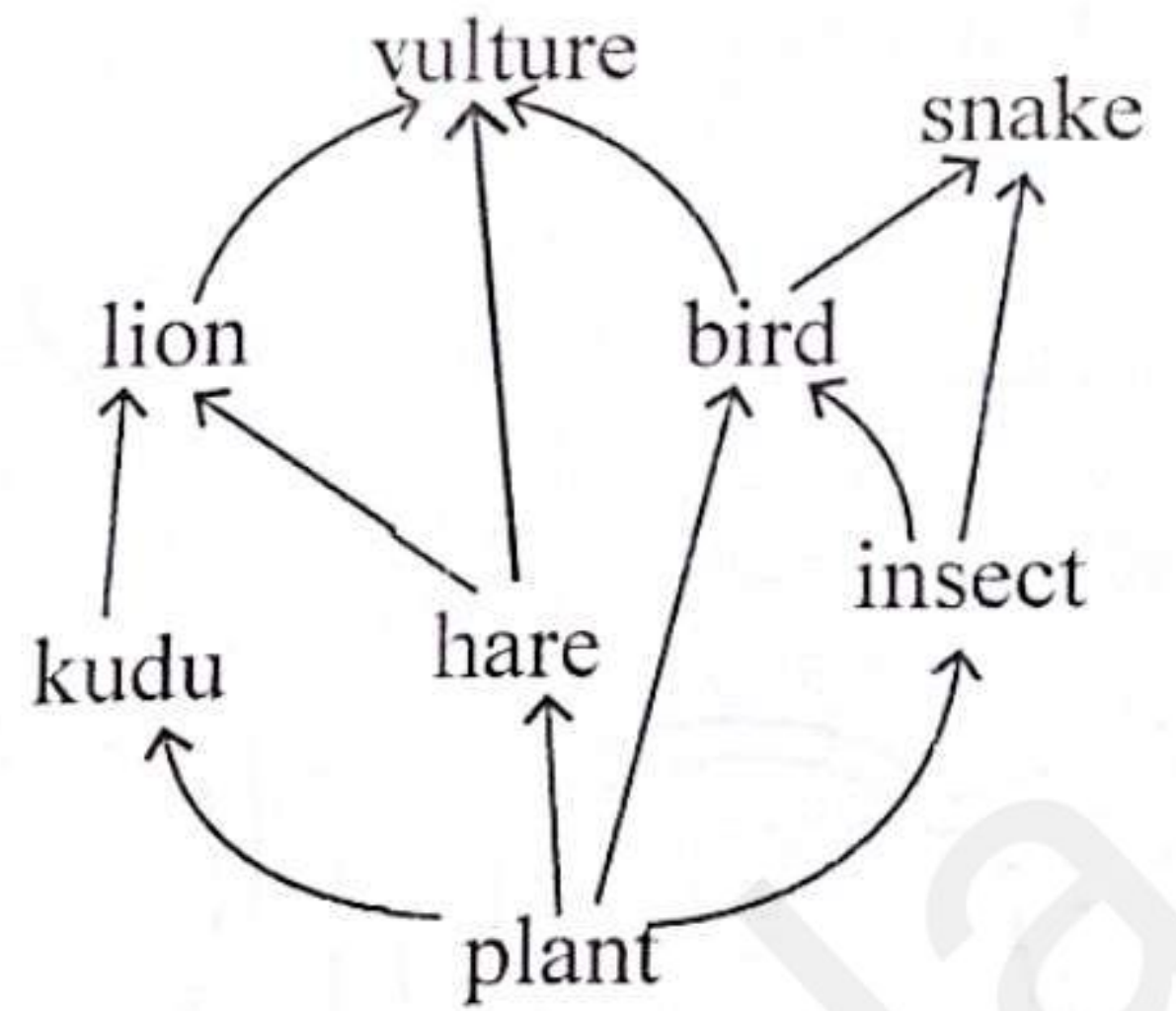
8. The diagram shows a leaf from a plant kept in the sun for 48 hours.



The leaf was tested for starch using iodine solution.
Which colour was observed on the green part?

- A green
- B white
- C brown
- D blue-black

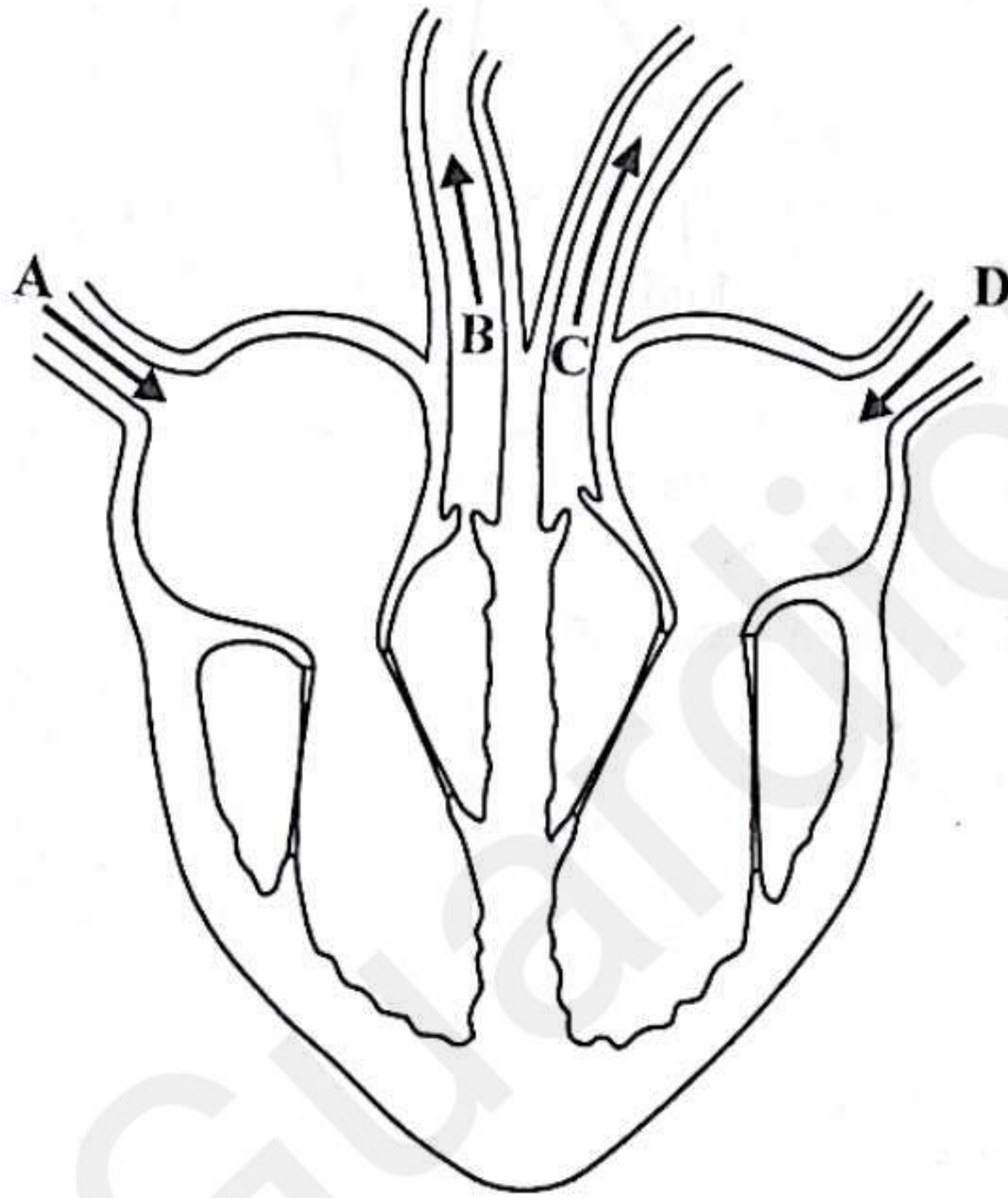
9. The diagram shows a food web.



Which organism is a primary consumer?

- A vulture
 - B snake
 - C hare
 - D plant
10. In the carbon cycle, which process releases carbon into the atmosphere?
- A respiration
 - B photosynthesis
 - C lightning
 - D feeding
11. Which component of blood is responsible for blood clotting?
- A plasma

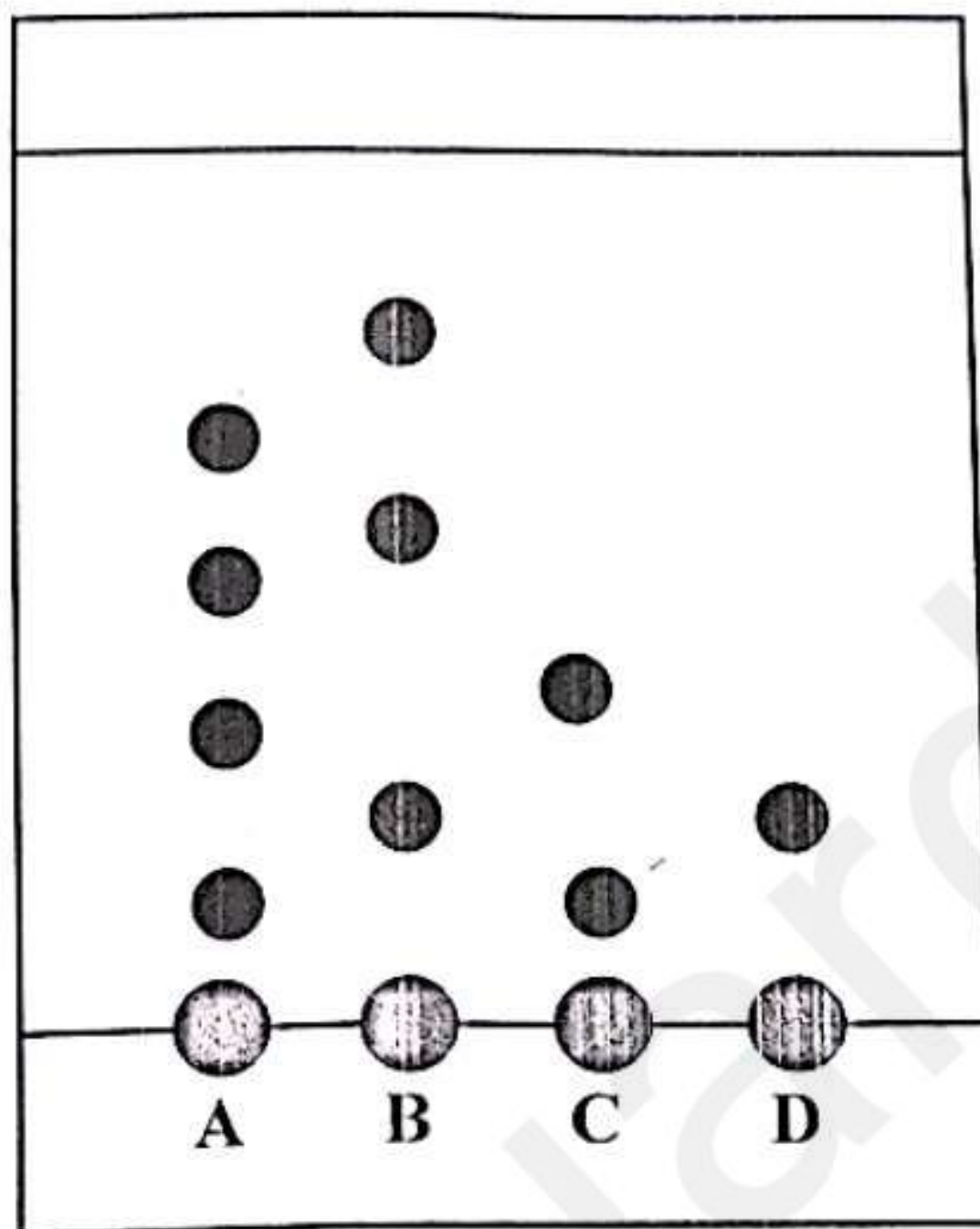
12. The diagram shows a vertical section through the human heart.
Which arrow, A, B, C or D, shows the movement of oxygenated blood from the lungs.



13. In which part of the female reproductive system does fertilisation take place?
- A ovary
 - B uterus
 - C oviduct
 - D vagina
14. A farmer planted X seeds and 30% germinated.
What is the value of X if 12 seeds germinated?
- A 21
 - B 40
 - C 60

15. The diagram shows results of a paper chromatography process for four samples of a chemical substance.

Which sample, A, B, C or D, of the chemical substance has two dyes?



16. Element **K** has an electronic configuration of 2, 8, 6 and element **P** has an electronic configuration of 2, 8, 2.
- Which one is the correct chemical formula of the product of the reaction between **K** and **P**?
- A KP
 - B K_2P
 - C KP_2
 - D K_2P_3
17. What is the proton number of a sodium atom, $^{23}_{11}\text{Na}$?
- A 11
 - B 12
 - C 22

18. What is the molecular mass of Ca(OH)_2 ?
[Ar of Ca = 40.0, Ar of O = 16.0 and Ar of H = 1.0]

- A 58.0
- B 62.0
- C 74.0
- D 124.0

19. It took 3 minutes for 5.0 g of copper sulphate granules to completely dissolve in 50 cm^3 of cold water.

Which condition decreases the time taken for the copper sulphate to completely dissolve?

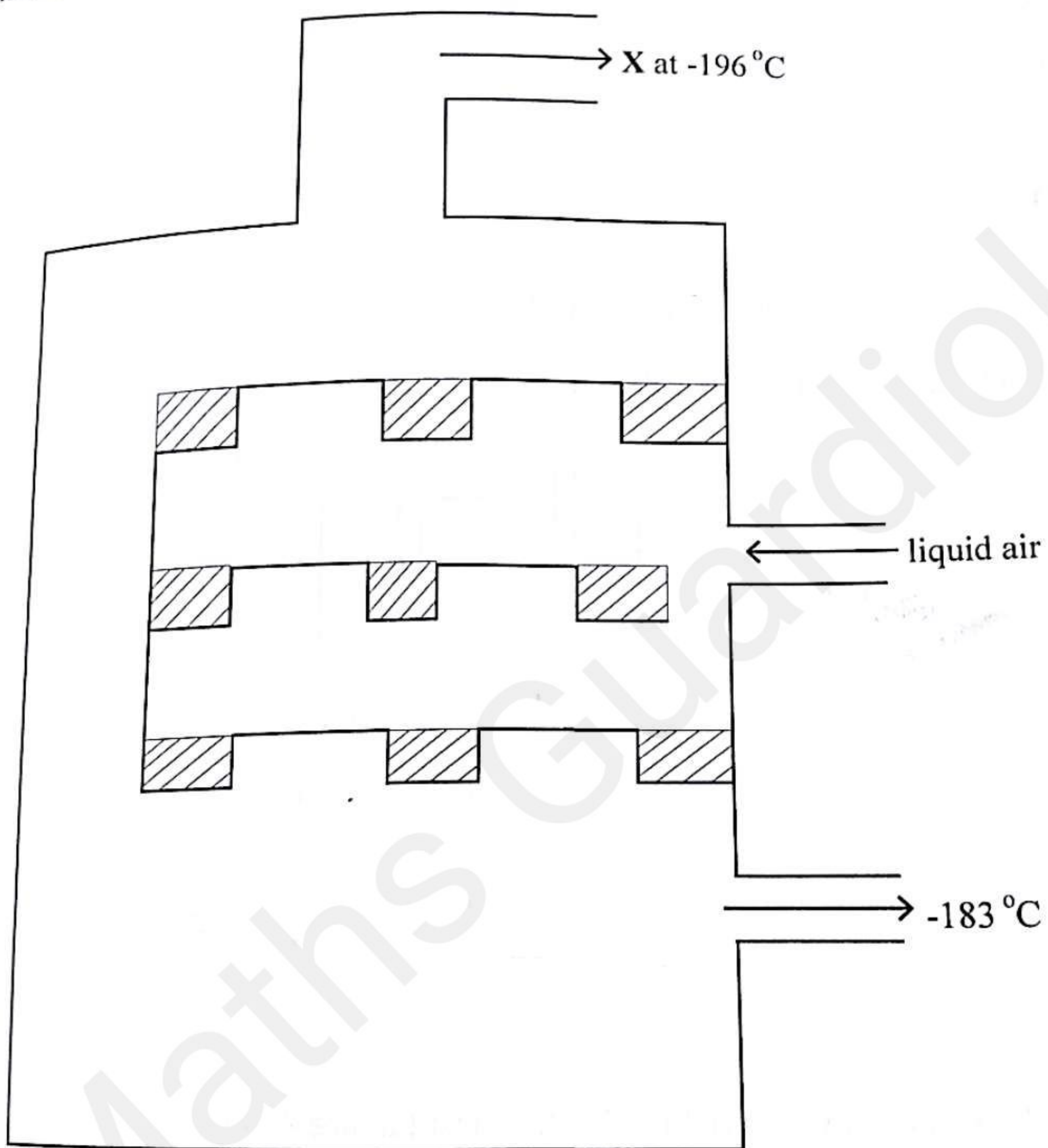
- A decreasing the volume of the water
- B decreasing the temperature of the water
- C increasing the temperature of the water
- D increasing the size of the copper sulphate granules

20. The Table shows pH values of four solutions, A, B, C and D.

Which solution, A, B, C or D, is alkaline?

solution	pH
A	1
B	4
C	7
D	9

1. The diagram shows the fractional distillation of liquid air.



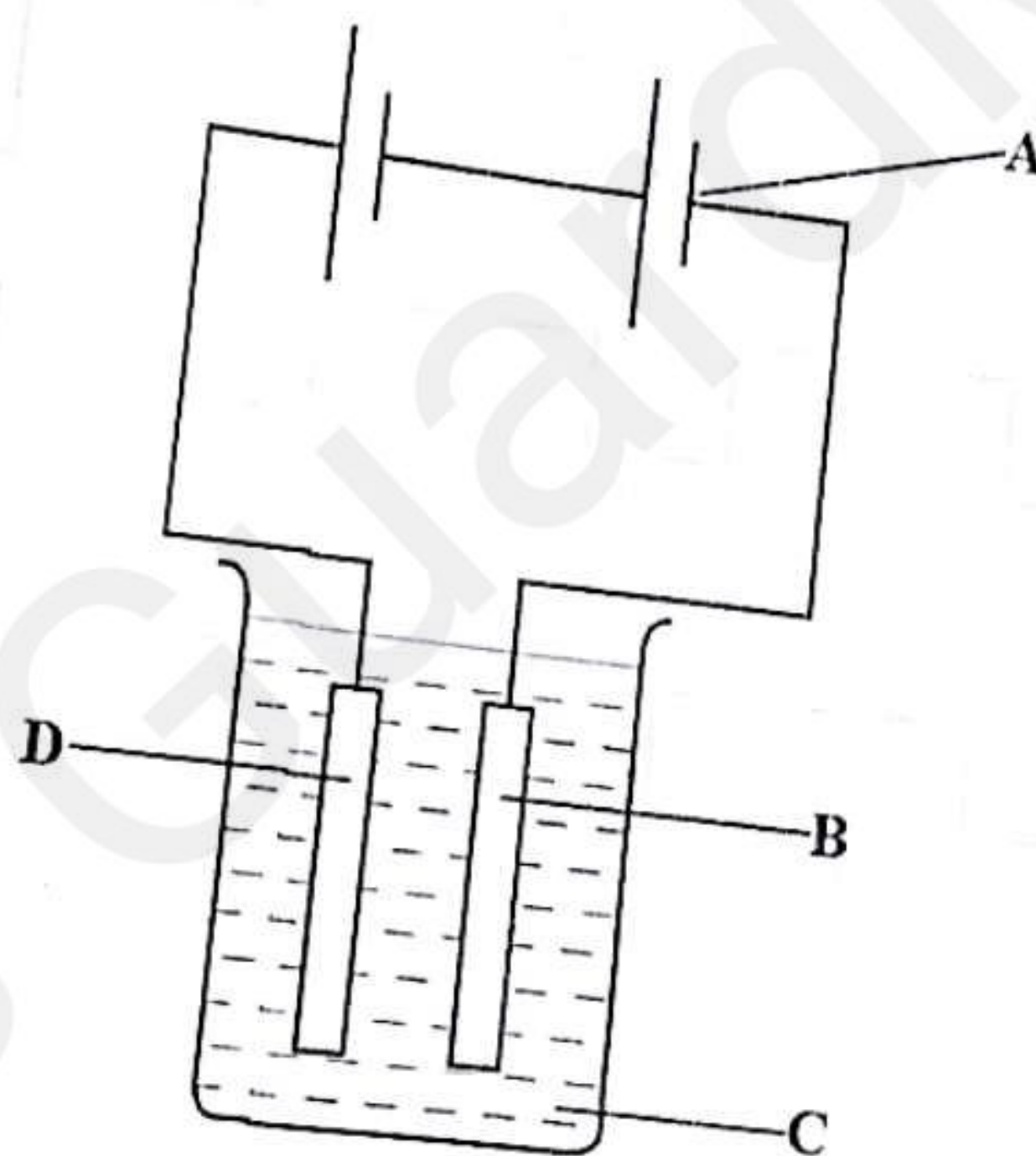
Which gas is collected at **X**?

- A oxygen
- B nitrogen
- C hydrogen

22. Which halogen is used to purify water?

- A bromine
- B fluorine
- C chlorine
- D iodine

23. The diagram shows an electrolytic cell.
Which part, A, B, C or D, is the anode?



24. Which one is a raw material used in the blast furnace during the extraction of iron?

- A coke
- B slag
- C cold air
- D carbon monoxide

Which catalyst is used in the manufacture of sulphuric acid?

- A iron
- B platinum
- C rhodium
- D vanadium (V) oxide

Which one is a product of the complete combustion of methane, CH_4 ?

- A oxygen
- B carbon soot
- C carbon dioxide
- D carbon monoxide

27. A gas is produced during the fermentation of glucose.

How is the gas identified?

- A it relights a glowing splint
- B it changes limewater from clear to milky
- C it gives a pop sound with a burning splint
- D it changes cobalt chloride paper from blue to pink

28. Which one is a form of potential energy?

- A heat energy
- B sound energy
- C kinetic energy
- D chemical energy

9. Which statement, on magnetism, is correct?

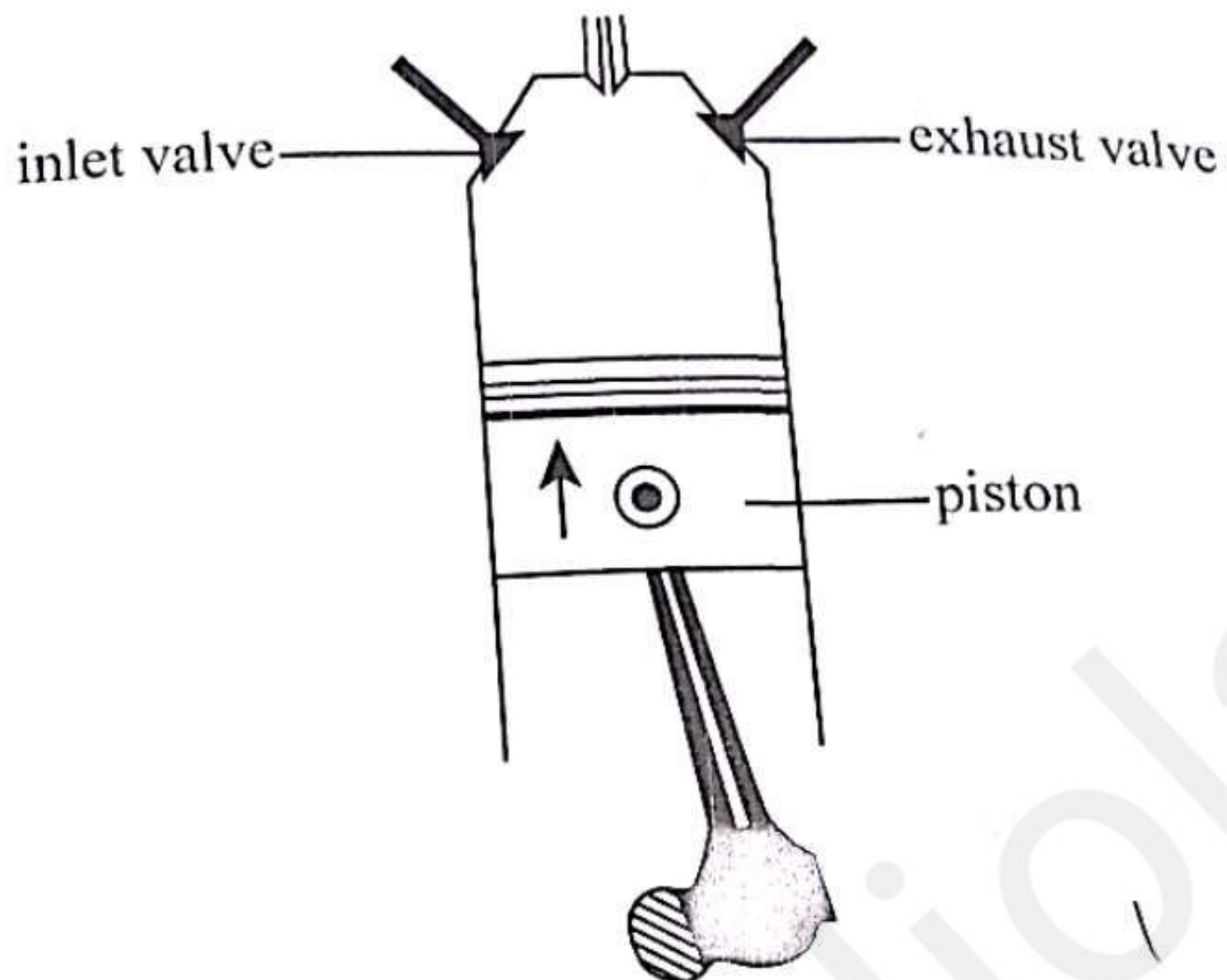
- A All metals can be magnetised.
- B All non-metals can be magnetised.
- C Ferromagnetic materials can be magnetised.
- D Any electrical conductor can be magnetised.

An irregular stone of mass 80.0 g was lowered into a measuring cylinder containing water. The level of the water in the measuring cylinder rose from 16.0 cm³ to 20.0 cm³. What is the density of the stone?

- A 0.05 g/cm³
- B 4.0 g/cm³
- C 5.0 g/cm³
- D 20.0 g/cm³

31.

The diagram shows a stroke of an engine.



Which stroke is shown?

- A compression
- B power
- C intake
- D exhaust

32.

What is inertia?

- A the weight of a body
- B the momentum of a body
- C the body's reaction to external force
- D the reluctance of a body to change its state of rest or its velocity

33.

The speed of an electric motor can be increased by

- A increasing the resistance of the coil.

- B** 25.00 Ω ✓
C 104.00 Ω
D 400.00 Ω

35. What is the unit for work done?

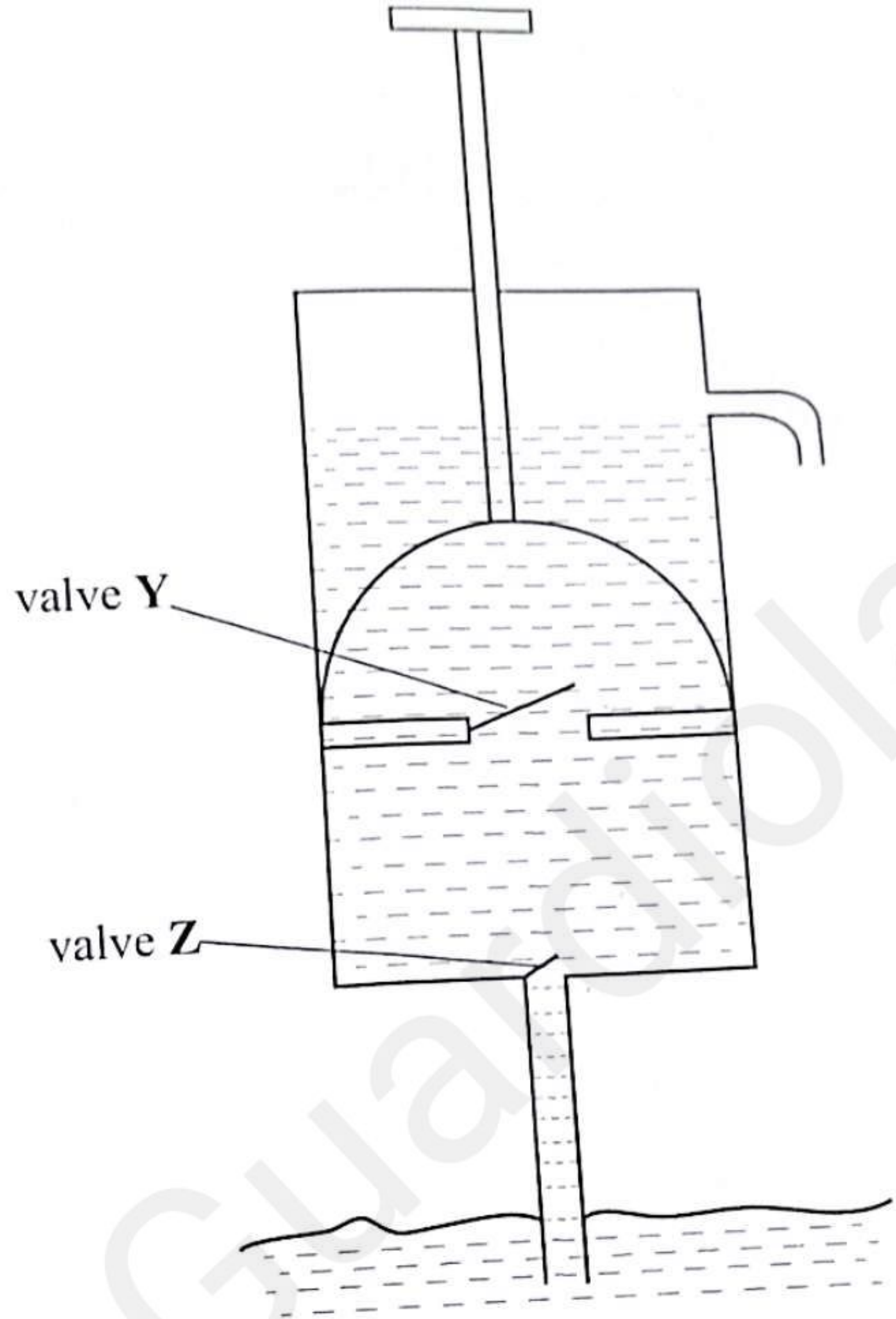
- A** watt
B ohm
C joule
D newton

The unit pascal can be expressed as

- A** N/m
B N/m²
C m/N
D m²/N

A machine raises a load of 20 N using an effort of 5 N.
What is the mechanical advantage, MA, of the machine?

0.25



What happens to the valves **Y** and **Z** during the upward stroke?

- A valve **Z** opens, valve **Y** closes
- B valve **Y** opens, valve **Z** closes
- C both valves open
- D both valves close

39. Pipes of a solar water heater are painted black because

- A black is a good absorber of heat.
- B black is a good reflector of heat.

Energy is converted to other forms of energy when using a telephone?